

Overview of K-nearest neighbors algorithm

Subject: K-nearest neighbors algorithm

1.

Error rates There are many results on the error rate of the k nearest neighbour classifiers. The k -nearest neighbour classifier is strongly (that is for any joint distribution on (X, Y)) consistent provided $k := k_n$ diverges and k_n/n converges to zero as $n \rightarrow \infty$. Let C_{n,k_n} denote the k nearest neighbour classifier based on a training set of size n . Under certain regularity conditions, the excess risk yields the following asymptotic expansion

2.

Data reduction Data reduction is one of the most important problems for work with huge data sets. Usually, only some of the data points are needed for accurate classification. Those data are called the prototypes and can be found as follows:

3.

$$R(R(C_{n,w,n})) - R(R(C_{\text{Bayes}})) = (B_1 s_n^2 + B_2 t_n^2) \{1 + o(1)\},$$
$$\{R\}_{\text{mathcal}}(C_n^{w,n}) - \{R\}_{\text{mathcal}}(C_{\text{Bayes}}) = \left(B_1 s_n^2 + B_2 t_n^2 \right) \{1 + o(1)\},$$

4.

Compute the Euclidean or Mahalanobis distance from the query example to the labeled examples. Order the labeled examples by increasing distance. Find a heuristically optimal number k of nearest neighbors, based on RMSE. This is done using cross validation. Calculate an inverse distance weighted average with the k -nearest multivariate neighbors.

5.

A drawback of the basic "majority voting" classification occurs when the class distribution is skewed. That is, examples of a more frequent class tend to dominate the prediction of the new example, because they tend to be common among the k nearest neighbors due to their large number. One way to overcome this problem is to weight the classification, taking into account the distance from the test point to each of its k nearest neighbors. The class (or value, in regression problems) of each of the k nearest points is multiplied by a weight proportional to the inverse of the distance from that point to the test point. Another way to overcome skew is by abstraction in data representation. For example, in a self-organizing map (SOM), each node is a representative (a center) of a cluster of similar points, regardless of their density in the original training data. k -NN can then be applied to the SOM.

6.

Feature extraction When the input data to an algorithm is too large to be processed and it is suspected to be redundant (e.g. the same measurement in both feet and meters) then the input data will be transformed into a reduced representation set of features (also named features vector). Transforming the input data into the set of features is called feature extraction. If the features extracted are carefully

chosen it is expected that the features set will extract the relevant information from the input data in order to perform the desired task using this reduced representation instead of the full size input. Feature extraction is performed on raw data prior to applying k-NN algorithm on the transformed data in feature space. An example of a typical computer vision computation pipeline for face recognition using k-NN including feature extraction and dimension reduction pre-processing steps (usually implemented with OpenCV):

7.

k-NN outlier The distance to the k th nearest neighbor can also be seen as a local density estimate and thus is also a popular outlier score in anomaly detection. The larger the distance to the k -NN, the lower the local density, the more likely the query point is an outlier. Although quite simple, this outlier model, along with another classic data mining method, local outlier factor, works quite well also in comparison to more recent and more complex approaches, according to a large scale experimental analysis.

8.

Further reading Dasarthy, Belur V., ed. (1991). Nearest Neighbor (NN) Norms: NN Pattern Classification Techniques. IEEE Computer Society Press. ISBN 978-0818689307. Shakhnarovich, Gregory; Darrell, Trevor; Indyk, Piotr, eds. (2005). Nearest-Neighbor Methods in Learning and Vision. MIT Press. ISBN 978-0262195478.

9.

Select the class-outliers, that is, training data that are classified incorrectly by k-NN (for a given k) Separate the rest of the data into two sets: (i) the prototypes that are used for the classification decisions and (ii) the absorbed points that can be correctly classified by k-NN using prototypes. The absorbed points can then be removed from the training set.

10.

$w_{ni} = 0$ for $i = k + 1, \dots, n$. With optimal weights the dominant term in the asymptotic expansion of the excess risk is $O(n^{-\frac{4}{d+4}})$. Similar results are true when using a bagged nearest neighbour classifier.

11.

The training examples are vectors in a multidimensional feature space, each with a class label. The training phase of the algorithm consists only of storing the feature vectors and class labels of the training samples. In the classification phase, k is a user-defined constant, and an unlabeled vector (a query or test point) is classified by assigning the label which is most frequent among the k training samples nearest to that query point.

12.

Parameter selection The best choice of k depends upon the data; generally, larger values of k reduces effect of the noise on the classification, but make boundaries between classes less distinct. A good k can be selected by various heuristic techniques (see hyperparameter optimization). The special case

where the class is predicted to be the class of the closest training sample (i.e. when $k = 1$) is called the nearest neighbor algorithm. The accuracy of the k -NN algorithm can be severely degraded by the presence of noisy or irrelevant features, or if the feature scales are not consistent with their importance. Much research effort has been put into selecting or scaling features to improve classification. A particularly popular approach is the use of evolutionary algorithms to optimize feature scaling. Another popular approach is to scale features by the mutual information of the training data with the training classes. In binary (two class) classification problems, it is helpful to choose k to be an odd number as this avoids tied votes. One popular way of choosing the empirically optimal k in this setting is via bootstrap method.